BAYESIAN MODELING OF COMPLEXITY

Bayesian Hierarchical Models I

Meysam Goodarzi
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Hertie School of Public Governance

Agenda

1 Recap & Today

2 Bayesian Hierarchical Models

- Introduction
- Fundamental Structure
- Multi-level Linear Models
- Nested vs. Non-Nested

3 Applications

4 Wrap-up & Next

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Course Progress

Where we've been:

- Sessions 1-3: Complexity and ABMs
- Session 4-5: Probability theory and Bayesian statistics (conjugate priors)
- Session 6: MCMC (sampling from posteriors)
- Session 7: Bayesian calibration
- Session 8: Bayesian linear models (regression)
- Session 9: Model assessment and sensitivity analysis

Today: Hierarchical models

Hierarchical models matter because:

- Most real data has structure (students in schools, agents in groups)
- Borrowing strength across groups (partial pooling)

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Why Hierarchical Models?

Real-world data has structure:

- Data is recorded and aggregated on **multiple levels** at **different times**
- Levels need to be incorporated in models
- Correlations can be identified by recognizing hierarchies
- Ignoring structure leads to wrong conclusions

Hierarchical models:

- Data conditioned on params
- Parameters conditioned on hyperparameters
- ... (can continue deeper)
- Natural Bayesian framework

Key advantage:
Borrowing strength across groups without assuming groups are identical

→ Since unknown parameters are specified with distributions, hierarchical models are essentially Bayesian.

Example: Radon in Homes

Problem: Measure radon levels in homes in Minnesota counties

- 85 counties
- Multiple homes per county (ranging from 1 to 100+)
- Some counties have lots of data, others very few

Question: How to estimate the radon level in each county?

Option 1: Complete pooling

- Ignore counties entirely
- One estimate for all of Minnesota

→ Loses county information

Option 2: No pooling

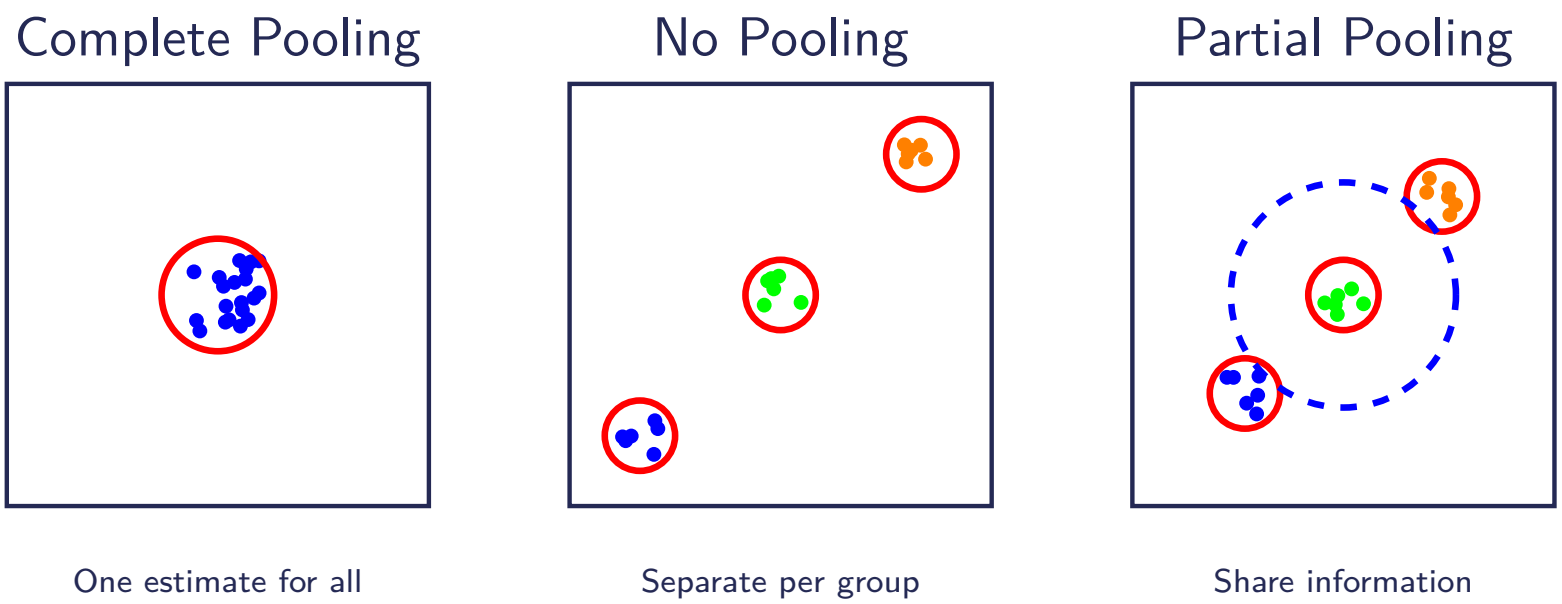
- Separate estimate per county
- No information sharing

→ Poor estimates for small counties

Option 3: Partial pooling (hierarchical!)

- Counties share information through common distribution
- Small counties borrow strength from larger ones
- Balance between extremes → Best of both worlds!

Visual Intuition: Pooling Strategies



Partial pooling: Groups shrink toward global mean, but maintain distinctiveness

Examples of Hierarchical Data

Common structures requiring hierarchical models:

Social Sciences:

- Students within schools
- Schools within districts

Economics:

- Employees within firms
- Firms within industries

ABMs:

- Agents within neighborhoods
- Neighborhoods within cities
- Cities within regions
- Regions within countries

Key insight: If you can say "units *within* groups", you need hierarchical models!

Structure of Hierarchical Models

Recall the standard Bayesian inference:

$$P(\theta|\mathcal{D}) \propto P(\mathcal{D}|\theta)P(\theta)$$

What if $P(\theta)$ depends on other parameters ψ?

We can write:

$$P(\theta|\mathcal{D}) \propto P(\mathcal{D}|\theta)P(\theta|\psi)P(\psi)$$

Components:

- $P(\mathcal{D}|\theta)$: Likelihood (data model)
- $P(\theta|\psi)$: Prior (depends on hyperparameters)
- $P(\psi)$: **Hyperprior** (prior on hyperparameters)

The chain can continue: $P(\psi)$ might itself depend on ϕ, etc.

Key question: When to stop adding levels?

Simple Example: Normal Hierarchy

Example: Heights of students in different schools. **The model** is:

Data level:	$X_{ij} \sim \mathcal{N}(\mu_j, \sigma_0^2)$	(student i in school j)
Group level:	$\mu_j \sim \mathcal{N}(m_\mu, s_\mu^2)$	(school means)
Hyperpriors:	$m_\mu \sim \mathcal{N}(170, 20^2)$	(global mean)
	$s_\mu \sim \text{HalfCauchy}(5)$	(between-school variation)

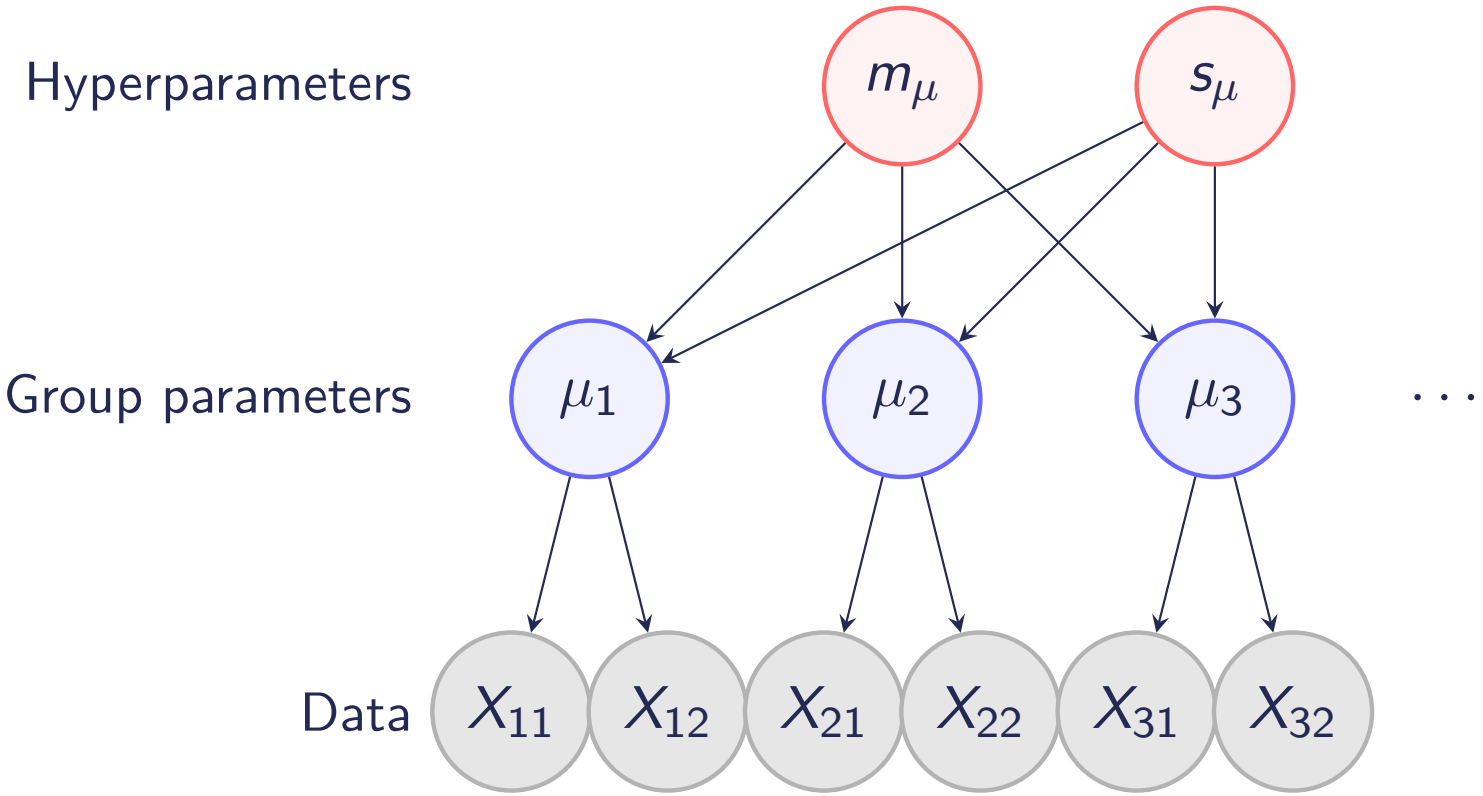
Interpretation:

- Each school j has its own mean height μ_j
- School means are NOT independent, i.e., they come from a common distribution $\mathcal{N}(m_\mu, s_\mu^2)$
- **Partial pooling**: schools share information through (m_μ, s_μ)

Benefits:

- Small schools borrow strength from the global distribution
- Large schools mostly use their own data
- Automatic shrinkage based on sample size!

Graphical Representation



Information flows both ways:

- Data informs group parameters ($X_{ij} \to \mu_j$)
- Group parameters inform hyperparameters ($\mu_j \to m_\mu, s_\mu$)
- Hyperparameters inform all groups (partial pooling!)

Priors & Hyperpriors

How far can we continue adding levels?

- Usually 2-3 levels sufficient for most problems
- Add levels only if substantively meaningful
- More levels = more complexity, harder inference

When to stop?

1. **Weakly informative priors:** Priors have minimal impact
2. **Domain knowledge:** We can directly specify a reasonable prior
3. **Computational limits:** MCMC becomes too slow
4. **Data limitations:** There are not enough groups to estimate hyperparameters

Rule of thumb:

- Need at least 5-10 groups to estimate between-group variation
- With fewer groups, use fixed hyperparameters (empirical Bayes)
- With many groups (>20), full hierarchical model works well

Multi-level Linear Models

The basic multi-level linear model: $Y_{ij} = \beta_{j0} + \beta_{j1} X_{ij} + \epsilon_{ij}$ with group index $j = 1, \ldots, J$.

Add second level that explicitly nests effects within groups:

$$\beta_{j0} = \gamma_{00} + \gamma_{10} Z_j + u_{j0},$$
$$\beta_{j1} = \gamma_{01} + \gamma_{11} Z_j + u_{j1}$$

Components:

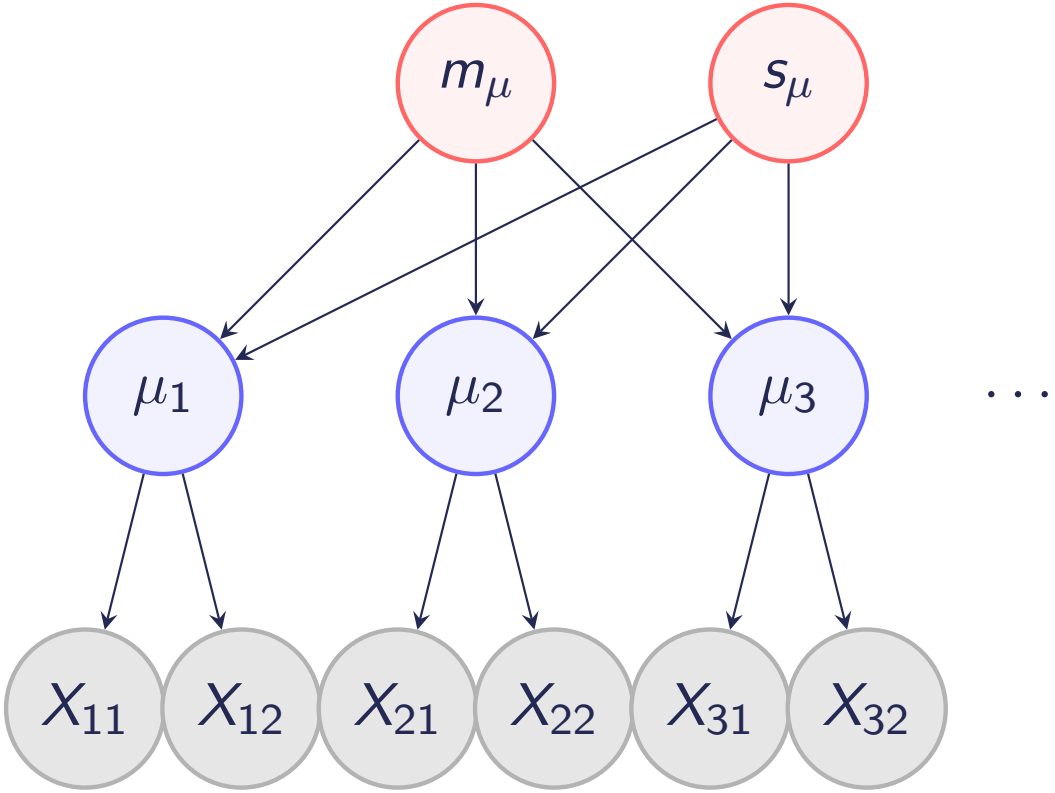
- Y_{ij}: Outcome for individual i in group j
- X_{ij}: Individual-level predictor
- Z_j: Group-level predictor (context variable)
- u_{j0}, u_{j1}: Random effects (group-level residuals)
- γ: Fixed effects (population-level parameters)

Key insight: Both intercepts AND slopes can vary by group!

Graphical Representation

$$\beta_{j0} = \gamma_{00} + \gamma_{10} Z_j + u_{j0}$$
$$\beta_{j1} = \gamma_{01} + \gamma_{11} Z_j + u_{j1}$$

$$Y_{ij} = \beta_{j0} + \beta_{j1} X_{ij} + \epsilon_{ij}$$



Combined Form

Substituting the second level into first:

$$\begin{aligned} Y_{ij} &= \beta_{j0} + \beta_{j1} X_{ij} + \epsilon_{ij} \\ &= (\gamma_{00} + \gamma_{10} Z_j + u_{j0}) + (\gamma_{01} + \gamma_{11} Z_j + u_{j1}) X_{ij} + \epsilon_{ij} \\ &= \gamma_{00} + \gamma_{01} X_{ij} + \gamma_{10} Z_j + \gamma_{11} X_{ij} Z_j + u_{j0} + u_{j1} X_{ij} + \epsilon_{ij} \end{aligned}$$

Composite error term: $u_{j1} X_{ij} + u_{j0} + \epsilon_{ij}$

Heteroskedasticity!

- Error variance depends on X_{ij}
- Standard OLS invalid
- Need hierarchical (or mixed-effects) approach

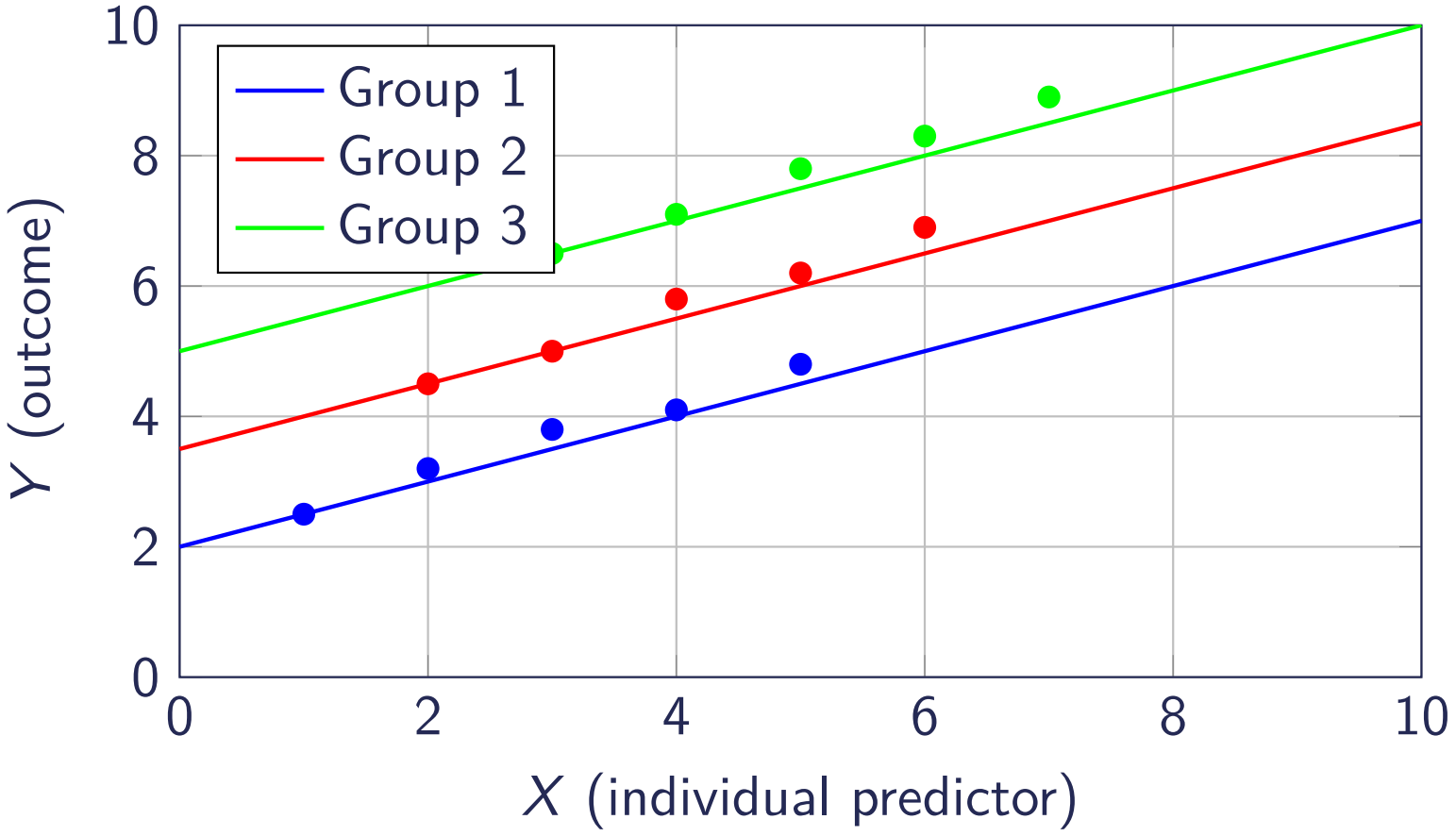
Cross-level interaction: $\gamma_{11} X_{ij} Z_j$

- Effect of individual-level X depends on group-level Z
- Example: Effect of study hours depends on school quality

Taxonomy of Multi-level Models

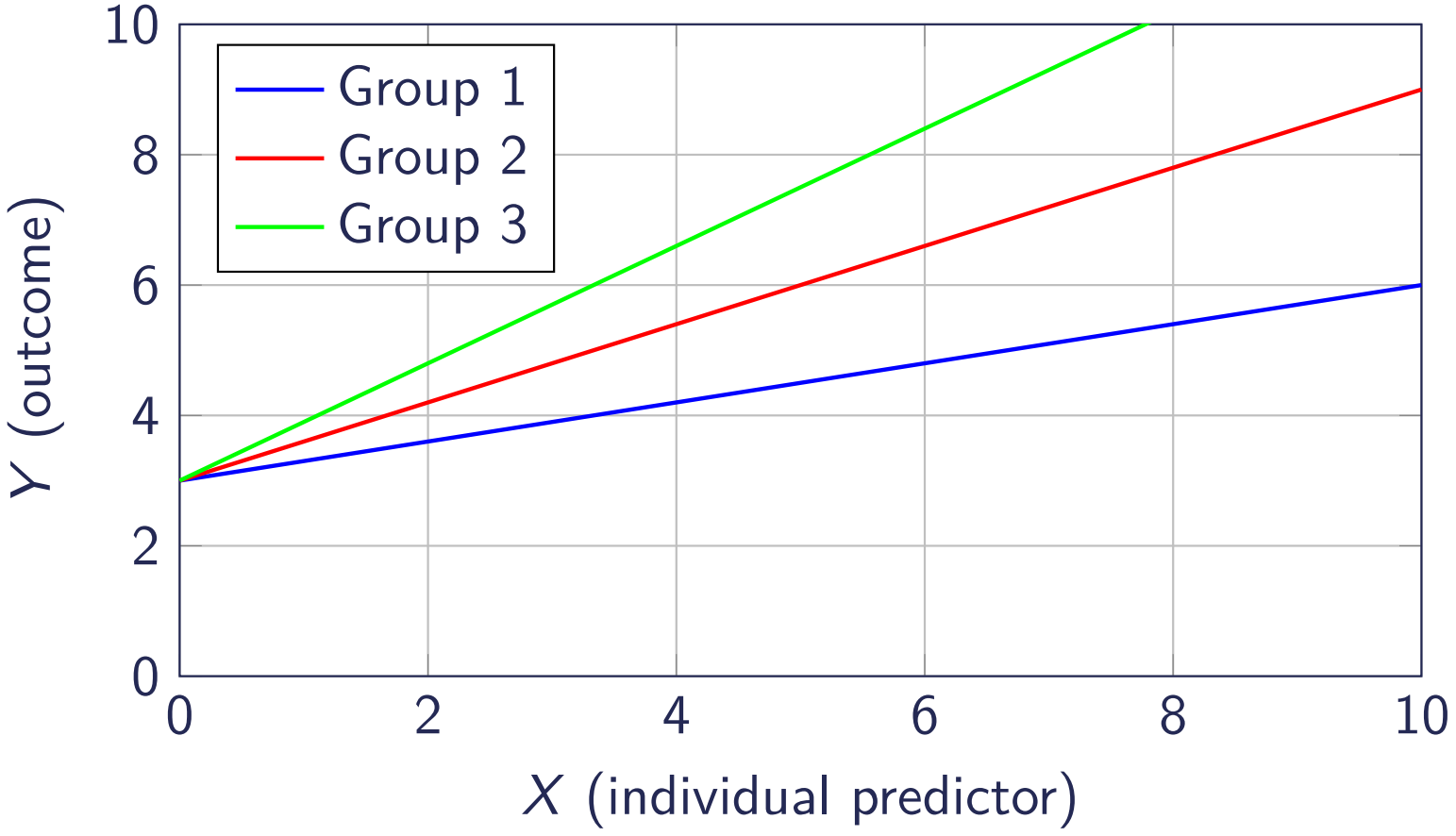
Name	**Model**
Fully Pooled	$Y_{ij} = \alpha + X_i'\beta + Z_j'\gamma + \epsilon_{ij}$
Random Intercept	$Y_{ij} = \alpha_j + X_{ij}'\beta + \epsilon_{ij}$
Random Slope	$Y_{ij} = \alpha + X_{ij}'\beta_j + Z_j'\gamma + \epsilon_{ij}$
Random Intercept-Random Slope	$Y_{ij} = \alpha_j + X_{ij}'\beta_j + Z_j'\gamma + \epsilon_{ij}$
Fully Unpooled	$Y_{i,j=1} = \alpha_{j=1} + X_{i,j=1}'\beta_{j=1} + Z_{j=1}'\gamma_{j=1} + \epsilon_{i,j=1}$ $Y_{i,j=2} = \alpha_{j=2} + X_{i,j=2}'\beta_{j=2} + Z_{j=2}'\gamma_{j=2} + \epsilon_{i,j=2}$ $\vdots$ $Y_{i,j=J} = \alpha_{j=J} + X_{i,j=J}'\beta_{j=J} + Z_{j=J}'\gamma_{j=J} + \epsilon_{i,j=J}$

Visual: Random Intercept Model



Random Intercept: Parallel lines (same slope β, different intercepts α_j)

Visual: Random Slope Model



Random Slope: Different slopes β_j, same intercept α

Note: Lines cross! Effect of X differs by group.

How Do We Know if Grouping Makes Sense?

Intraclass Correlation (ICC):

Proportion of total variance due to between-group differences:

$$\text{ICC} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}}$$

Interpretation:

- ≈ 0: No clustering, groups don't matter $\rightarrow$ standard regression
- ≈ 0.05-0.10: Modest clustering (hierarchical model helpful)
- > 0.10: Strong clustering $\rightarrow$ hierarchical model essential!

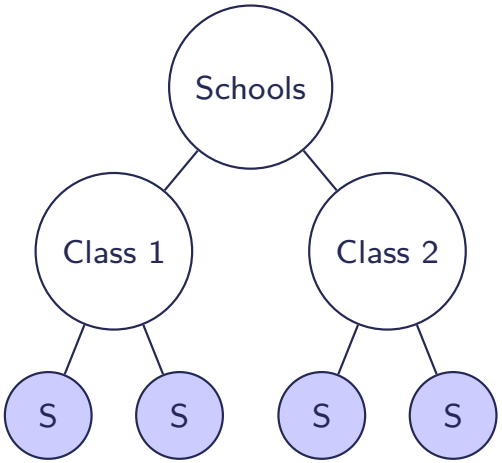
Example: Student test scores

- ICC $= 0.15$ means 15% of variance is between schools
- 85% is within schools (student-level)
- Ignoring school clustering underestimates standard errors

Nested vs. Non-Nested Hierarchies

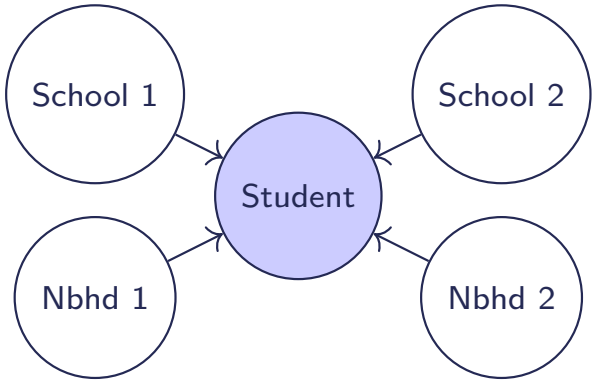
Nested (tree structure):

- Each lower level belongs to exactly one upper level
- Clear parent-child relationships
- students → classes → schools



Non-Nested (crossed):

- Lower levels can belong to multiple upper levels
- Network structure
- students → (schools & neighborhoods)



Key difference:

- Nested: One random effect per level
- Non-nested: Multiple random effects at the same level (can interact!)

Examples: Nested vs. Non-Nested

Nested examples:

- **Nested:** Each level depends only on one level above

$$y_i \sim \mathcal{P}(\lambda_i) \rightarrow \lambda_i \sim \mathcal{G}_\lambda(\alpha, \beta) \rightarrow \begin{cases} \alpha \sim \mathcal{G}_\alpha(A, B) \\ \beta \sim \mathcal{G}_\beta(C, D) \end{cases}$$

- Employees within departments within companies
- Cells within tissues within organisms

Non-nested examples:

- **Non-Nested:** Each level can depend on multiple units

$$y_i \sim \mathcal{P}(\lambda_i) \rightarrow \log(\lambda_i) = \mu + \alpha_{\text{school}[i]} + \beta_{\text{ngbh}[i]} \rightarrow \begin{cases} \alpha \sim \mathcal{N}(0, \sigma_\alpha^2) \\ \beta \sim \mathcal{N}(0, \sigma_\beta^2) \end{cases}$$

- Students crossed by schools and neighborhoods
- ABM agents influenced by multiple overlapping networks

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Student Math Performance

Data: Math scores from Italian middle school students

Structure: 1,350 students, 10 schools, varying sample sizes

- Schools 0 & 1: Small samples (n=10, 20) - artificially reduced
- Schools 2-9: Balanced samples (n=140-160)

Question: How does baseline math_t1 predict final math_t2?

Models to compare:

1. **Complete pooling:** $\text{math_t2}_i = \alpha + \beta \cdot \text{math_t1}_i + \epsilon_i$
2. **No pooling:** $\text{math_t2}_i = \alpha_{\text{school}[i]} + \beta_{\text{school}[i]} \cdot \text{math_t1}_i + \epsilon_i$
3. **Partial pooling (hierarchical - Random Intercepts):**

$$\begin{aligned} \text{math_t2}_i \sim \mathcal{N}(\mu_i, \sigma^2), \quad & \mu_i = \alpha_{\text{school}[i]} + \beta \cdot \text{math_t1}_i \\ & \alpha_j \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) \\ & \mu_\alpha \sim \mathcal{N}(0, 5), \quad \sigma_\alpha \sim \text{HalfCauchy}(2) \end{aligned}$$

Example I: Shrinkage Effects

What happens with partial pooling?

- **Large schools (n=140-160):**
  - Stay close to their sample estimates
  - Minimal shrinkage toward global mean
  - High confidence in school-specific parameters
- **Small schools (n=10, 20):**
  - Shrink strongly toward global mean
  - "Borrow strength" from larger schools
  - Reduces extreme/unreliable estimates
- **Shrinkage magnitude:**
  - School 0 (n=10): Large shrinkage
  - School 1 (n=20): Moderate shrinkage
  - Schools 2-9 (n=140-160): Minimal shrinkage

Key result: Shrinkage is *automatic* and proportional to sample size!

Radon Levels (Classic)

Data: Radon measurements in Minnesota homes

Structure: 919 homes, 85 counties, varying sample sizes per county (1-116)

Models to compare: 1. Complete pooling: $\log(\text{radon})_i = \alpha + \epsilon_i$ **2. No pooling:** $\log(\text{radon})_i = \alpha_{\text{county}[i]} + \epsilon_i$

3. Partial pooling (hierarchical):

$$\begin{aligned} \log(\text{radon})_i &\sim \mathcal{N}(\alpha_{\text{county}[i]}, \sigma^2) \\ \alpha_j &\sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) \\ \mu_\alpha &\sim \mathcal{N}(1, 1^2), \quad \sigma_\alpha \sim \text{HalfCauchy}(1) \end{aligned}$$

Key insight: the hierarchical model balances individual county data with overall pattern

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Summary

Key concepts from today:

- **Partial pooling:** Balance between complete pooling and no pooling
- **Shrinkage:** Small groups borrow strength from overall pattern
- **Random effects:** Parameters that vary by group
- **Hyperparameters:** Parameters of parameter distributions
- **ICC:** Quantifies importance of grouping
- **Nested vs. crossed:** Different dependency structures

Why hierarchical models matter:

- Real data has structure — ignoring it is wrong
- Automatic regularization (no overfitting)
- Predictions for new groups possible

Questions?

Next: Advanced Topics in Hierarchical Modeling!